Research on UAV Remote Sensing Monitoring of Lightweight Wheat Lodging Segmentation Network Using Multimodal Fusion (RGB+DSM)

Chenchen Yang
School of Information Engineering
Shandong Huayu University of Technology
Dezhou, Shandong, China

Qang Wang*
College of Information and Management Science
Henan Agricultural University
Zhengzhou, Henan, China
*17863943627@163.com

Xiaohong Duan
School of Information Engineering
Shandong Huayu University of Technology
Dezhou, Shandong, China

2026 7th International Conference on Computer Vision, Image and Deep Learning (CVIDL) | 979-8-3195-4465-0/26/$31.00 ©2026 IEEE | DOI: 10.1109/CVIDL70130.2026.11637617

***Abstract*: UAV remote sensing combined with deep learning–based semantic segmentation has become a mainstream technique for the intelligent monitoring of wheat lodging. Nevertheless, current approaches are prone to interference from field shadows, weeds, and ridges, leading to blurred segmentation boundaries, frequent false and missed detections, excessive model parameters, and difficulty in real-time deployment on airborne platforms. To tackle these issues, this study proposes a lightweight wheat lodging segmentation network, ML-LodgingNet, using RGB images and DSM data synchronously collected by UAVs at the wheat maturity stage. Experimental results demonstrate that the proposed network achieves Intersection over Union (IoU) values of 89.42% and 70.65% on simple and complex test subsets, respectively, with corresponding segmentation accuracies of 95.61% and 90.13%. The model contains only 1.26 M parameters and delivers a real-time inference speed of 19.3 FPS on edge devices. Compared with the single RGB modality, the RGB+DSM multimodal fusion strategy improves key evaluation indicators by 4.2%–9.1% on average, significantly enhancing lodging segmentation accuracy and boundary integrity in complex field environments. This work provides an effective algorithmic support for real-time UAV-borne monitoring of wheat lodging.**

Keywords: wheat lodging; multimodal fusion; RGB; DSM; lightweight network; semantic segmentation; UAV remote sensing

I. INTRODUCTION

Wheat is highly prone to lodging during the middle and late growth stages, mainly caused by adverse weather conditions, unreasonable planting density, and improper water and fertilizer management. Under normal climatic conditions, wheat lodging can lead to a yield loss of 10%–30%, and this loss can exceed 50% in severe lodging cases[1-2]. Moreover, lodging can easily induce crop diseases, reduce grain quality, and lower the efficiency of mechanical harvesting. Unmanned aerial vehicle (UAV) remote sensing has become a core data acquisition approach for crop disaster monitoring owing to its merits of high flexibility, efficiency, spatial resolution, low cost, and weak terrain constraints[3]. Combining UAV data with deep learning-based semantic segmentation enables pixel-level automatic extraction of lodged wheat regions[4][5]. Nevertheless, existing methods still suffer from obvious limitations. This study exploits the complementary properties of RGB texture features and DSM elevation features, and designs a dual-branch lightweight encoder as well as a cross-modal dynamic gating fusion module[6]. On this basis, we construct ML-LodgingNet, a specialized segmentation network for wheat lodging that achieves a good balance among high accuracy, lightweight architecture, and real-time inference, providing a novel technical solution for rapid UAV-based monitoring of wheat lodging.

II. MATERIALS AND METHODSTITLE

A. Study Area and Data Acquisition

The study area is located in the wheat cultivation region of Qingfeng County, Puyang City, Henan Province. It has a northern subtropical monsoon climate featured by synchronous rainfall and heat as well as sufficient sunshine, which makes it a typical large-scale and standardized wheat production area. Data collection was carried out during the wheat maturity stage. Images were acquired between 11:00 and 14:00 under clear and calm weather conditions with stable illumination. A DJI Mavic Air drone was used to collect RGB visible images and DSM elevation data synchronously, ensuring high imaging quality and stable flight attitude. The partially acquired images are shown in Figure 1.

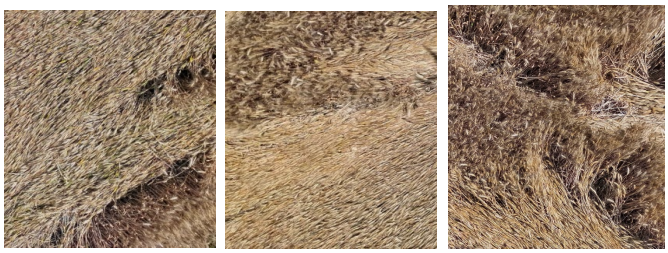


Figure 1. Sample image data of some lodged wheat

B. Data Preprocessing and Dataset Construction

To ensure the quality of raw multimodal data, eliminate errors caused by UAV flight attitude and imaging distortion, achieve precise alignment between RGB and DSM data, and provide standardized, high-quality data for model training, this study conducts systematic and standardized preprocessing on raw RGB imagery and DSM elevation data collected by UAVs. The core preprocessing workflow comprises five key stages: panoramic image stitching and geometric correction, pixel-level precise registration of multimodal data, uniform image

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:02:25 UTC from IEEE Xplore. Restrictions apply.

cropping, removal of invalid samples, and sample data augmentation.

Following preprocessing, a specialized multimodal dataset (RGB+DSM) for wheat lodging detection was constructed using the validated paired data. An efficient annotation strategy—involving uniform image cropping followed by precise sample annotation—was adopted to significantly reduce the workload and labor costs associated with large-scale, fine-grained, pixel-level field annotation. The professional image annotation tool LabelMe was utilized to perform precise pixel-level binary annotation, strictly categorizing areas into "wheat lodging target regions" and "non-lodging background regions"; specifically, red-annotated areas represent the core wheat lodging targets, while black-annotated areas represent non-lodging wheat and irrelevant background elements such as soil, weeds, and field ridges.

The final multimodal dataset contains a total of 10,460 valid samples, all consisting of paired RGB and DSM images. These samples were randomly split into training and testing sets at a standard 8:2 ratio. The training set comprises 8,000 samples, used primarily for iterative model training, optimization of network weights, and fine-tuning for model convergence; the testing set contains 2,460 samples, dedicated to the quantitative evaluation and performance testing of the model's segmentation capabilities.

C. ML-LodgingNet Network Architecture

The network adopts an encoder – decoder architecture, which consists of a dual-branch lightweight encoder, a cross-modal dynamic gating fusion module, and a lightweight decoder. This module firstly standardizes the channel dimensions and normalizes the semantic distribution of multi-scale feature maps output from the dual-branch encoder, which effectively eliminates fusion biases induced by the inconsistent channel numbers and feature value distributions between RGB and DSM modal features. Subsequently, a gated weight generation sub-network is constructed to generate adaptive fusion weight coefficients for RGB texture features and DSM elevation features, respectively. The two sets of weight coefficients are constrained to a sum of 1, which guarantees the balance and stability of multimodal feature fusion. Finally, the gated weighted multimodal features are deeply fused through channel concatenation. This operation significantly enhances the feature response intensity of texture and elevation information in wheat lodging regions, while suppressing the interference responses from irrelevant background features including field shadows, weeds, and field ridges. The mathematical formulation of the proposed cross-modal dynamic gating fusion strategy is defined as follows:

$$\mathbf{F}_{fuse} = \alpha \cdot \mathbf{F}_{rgb} + (1 - \alpha) \cdot \mathbf{F}_{dsm} \quad (1)$$

where $\mathbf{F}_{fuse}$ denotes the final fused multimodal feature map, $\mathbf{F}_{rgb}$ represents the texture feature map extracted from the RGB branch, $\mathbf{F}_{dsm}$ refers to the elevation structural feature map obtained from the DSM branch, and α is the adaptive gating weight coefficient learned end-to-end by the network, which is constrained within the range of [0, 1].

Figure 2 shows a detailed structural diagram of the gating fusion module.

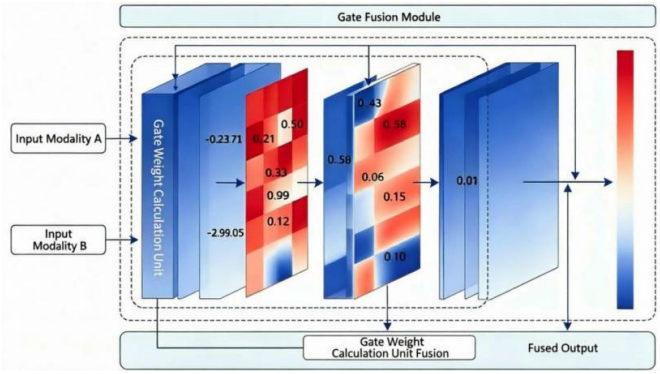


Figure 2.Detailed Structure Diagram of the Gated Fusion Module.The heatmap displays normalized gating weights, where values closer to 1 indicate greater reliance on Input Modality A, and values closer to 0 indicate preference for Modality B.This confirmsthe module's ability to adaptively selectthe optimalmodality.

1.The dual-branch lightweight encoder comprises an RGB texture branch and a DSM elevation branch with a symmetric structure and independent parameter updates. It adopts depthwise separable convolutions in the whole network, which reduces the number of parameters by more than 75% compared with standard convolutions. A lightweight cross-attention module is embedded at each downsampling stage to enhance lodging target features and suppress background noise.

2.The cross-modal dynamic gating fusion module performs channel unification and standardization for multi-scale features.

3.The lightweight decoder adopts bilinear interpolation upsampling and 1×1 convolutions to reduce computational cost. It fuses shallow detailed features via skip connections to alleviate edge blurring. Finally, pixel-level binary classification segmentation results are generated after the Sigmoid activation function..

D. Loss Function and Evaluation Metrics

In this study, a weighted composite loss function that combines Focal Loss and Dice Loss is designed according to the specific requirements of multimodal lightweight wheat lodging segmentation, so as to balance sample distribution and improve segmentation overlap accuracy[7]. The composite loss function is formulated as follows:

$$L_{total} = \lambda L_{focal} + (1 - \lambda) L_{dice} \quad (2)$$

Where: L_{total} denotes the total composite loss; L_{focal} is the focal loss; L_{dice} is the Dice loss; and λ is the balancing coefficient between the two loss terms, which is set to 0.5 in all experiments. The proposed composite loss function integrates the merits of both loss functions, which effectively improves the segmentation performance of the model in complex field scenarios, accelerates training convergence, and promotes the optimization of network parameters[8].

To comprehensively, objectively, and quantitatively evaluate the segmentation accuracy, lightweight feasibility, and real-time inference performance of the proposed ML-

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:02:25 UTC from IEEE Xplore. Restrictions apply.

LodgingNet network, three widely used metrics are adopted for segmentation accuracy: overall accuracy, F1-score, and Intersection over Union (IoU)[9]. Higher values of these three metrics reflect better segmentation precision and integrity. For lightweight and real-time performance evaluation, the number of parameters (Million) and edge inference frame rate (FPS) are utilized. A smaller parameter size indicates stronger model compression ability and less storage occupation, while a higher FPS represents faster inference speed and better adaptability for real-time monitoring on edge devices.[10] All metrics are evaluated in concert to comprehensively verify the model's overall application performance.

III. Results and Analysis

A. Test Environment

All experiments were implemented on the Windows Server 2022 operating system. The PyTorch deep learning framework (open-source) and Python 3.10 were employed for model development and training. Uniform standard hyperparameters were used throughout training: the batch size was set to 8, the initial learning rate to 0.001, the weight decay coefficient to 0.0001, and the total training epochs to 80[11]. The AdamW optimizer was adopted, and all hyperparameters were fixed during training to ensure consistent experimental settings, reliable results, and valid comparisons[12].

B. Comparative Experiments of Different Models

To comprehensively quantify and validate the superior performance and suitability of the proposed ML-LodgingNet—a multimodal lightweight segmentation network—for wheat lodging segmentation, four mainstream semantic segmentation models (UNet, DeepLabv3+, U2NetP, and SegNet) were selected as benchmarks. All models utilized the same RGB+DSM multimodal dataset, unified training hyperparameters, and identical experimental hardware and software environments. Iterative training and performance testing were conducted concurrently, with comprehensive comparative experiments performed on both simple and complex test subsets to quantitatively evaluate segmentation accuracy, model lightweightness, and real-time inference performance.[13] The comparison results regarding segmentation performance and lightweightness metrics for the different models are presented in Table 1.

TABLE 1. COMPARISON OF SEGMENTATION PERFORMANCE AND LIGHTWEIGHT METRICS OF DIFFERENT MODELS

Network model name	Simple test set IoU/ %	Complex test set IoU/%	Segmentation accuracy / %	Model parameter count / M	Edge inference frame rate/FPS
UNet	82.15	62.37	92.04	29.44	6.2
DeepLabv3+	83.57	64.12	92.87	36.72	4.8
U2NetP	85.23	66.89	93.45	1.13	15.6
SegNet	81.06	61.54	91.62	31.03	5.4
ML-LodgingNet (This paper)	89.42	70.65	95.61	1.26	19.3

The quantitative experimental results demonstrate that ML-LodgingNet outperforms all baseline models in segmentation accuracy, lightweight performance, and real-time inference speed. Its advantages are especially significant in complex field scenarios, fully satisfying the requirements for edge deployment on UAV platforms.

C. Ablation Test of Core Module

To quantitatively verify the effectiveness and performance contributions of the three core innovations, namely DSM elevation modality supplementation, multimodal fusion strategy, and cross-modal dynamic gating fusion module, three groups of controlled ablation experiments were conducted in this study. Following the single-variable principle, core modules were sequentially removed for comparative testing to accurately quantify the performance improvement brought by each module. The ablation results are presented in Table 2.

TABLE 2. ABLATION TEST RESULTS OF CORE MODULE

Ablation experimental group	RGB modal input	DSM Modal Input	Dynamic Gating Fusion Module	Test set IoU/%	Model parameter count / M
Experimental Group 1 (Single RGB mode only)	√	×	×	81.62	1.22
Experimental Group 2 (Simple splicing and fusion of RGB+DSM)	√	√	×	84.92	1.24
Experimental Group 3 (Complete Model in this Paper)	√	√	√	89.42	1.26

Quantitative results of the ablation study verify that all three core innovative modules are indispensable and contribute substantially to boosting the segmentation performance of the proposed model. When only RGB imagery is fed into the network, the model achieves the minimal IoU of merely 81.62%, accompanied by severe segmentation artifacts on complex and ambiguous field samples. Such a performance degradation reveals that unimodal RGB texture features exhibit poor anti-interference capability and fail to support accurate segmentation of lodged wheat under complex farmland scenarios, which highlights the necessity of introducing DSM elevation information as complementary modal data.Simple fixed-weight concatenation fusion that incorporates RGB and DSM features yields a 3.3% IoU improvement, which validates the superiority of multimodal information complementarity for segmentation accuracy, yet this straightforward fusion strategy still suffers from limited representation capacity. By embedding the cross-modal dynamic gated fusion module proposed in this work, the IoU metric is further elevated by 4.5%, delivering the optimal segmentation performance across all ablation variants. This substantial gain demonstrates that the presented adaptive dynamic gating mechanism mitigates the semantic discrepancy between different modalities, generating far more discriminative fused features compared with conventional naive concatenation fusion schemes.

D. Robustness Test

To fully verify the anti-interference capability and operational stability of the ML-LodgingNet network in

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:02:25 UTC from IEEE Xplore. Restrictions apply.

complex UAV field scenarios, and to reflect real-world conditions including varying illumination, crop occlusion, and image blurring, this study conducted dedicated robustness tests under three typical field disturbances: light intensity variation, crop occlusion, and image motion blur. These tests quantitatively evaluate the segmentation performance of the model under complex interference. The robustness test results are presented in Table 3.

TABLE 3. ROBUSTNESS TEST RESULTS OF THE MODEL IN COMPLEX SCENARIOS

Types of field interference conditions	Model segmentation accuracy / %
Normal standard operating conditions	95.61
Light intensity disturbance conditions	86.12
Field crop shading conditions	94.35
Image motion blur condition	90.07

Experimental data from robustness tests indicate that, although the segmentation accuracy of the proposed model drops slightly under various complex field interference conditions, it maintains a high overall level of precision with minimal performance fluctuation; the model demonstrates stable and reliable operation, making it suitable for actual field missions involving unmanned aerial vehicles.

IV. CONCLUSION

In this study, a multimodal dataset dedicated to wheat lodging is constructed using RGB images and DSM elevation data synchronously collected by UAVs. A lightweight semantic segmentation network with dual-branch encoding and dynamic gating fusion, namely ML-LodgingNet, is innovatively proposed. The segmentation performance, effectiveness of key modules, and field applicability of the model are comprehensively verified through comparative experiments, ablation experiments, and robustness tests. The core conclusions are drawn as follows

(1)RGB texture features and DSM elevation features are highly complementary. The RGB4DSM multimodal fusion strategy can effectively suppress interferences such as shadows and weeds, improving the core indicators of lodging segmentation by 4.2-9.1% on average, with more significant improvements observed in complex scenarios

(2)The cross-modal dynamic gating fusion module is able to adaptively bridge the semantic gap and dynamically assign feature weights. Its fusion performance is significantly superior to the traditional simple concatenation method, which constitutes the key to enhancing segmentation accuracy,

(3)ML-LodgingNet has only 1.26 M parameters and achieves an edge inference speed of 19.3 FPS. The IoU values on the simple and complex test sets reach 89.42%and 7065%, respectively. The model realizes a favorable balance between high precision and lightweight characteristics, making it suitable for real-time UAV-borne deployment

(4)The model exhibits strong robustness against illumination variations, crop occlusion, and motion blur, as well as satisfactory generalization ability. It can satisfy the demand for real-time UAV monitoring of field wheat lodging and provides a feasible technical approach for disaster monitoring in smart agriculture.

ACKNOWLEDGMENT

This work was supported by Dezhou Big Data and Intelligent Sensing Technology Engineering Research Center under the projects No. PT2025KJX002.

REFERENCES

[1] Azizi, A., Zhang, Z., Rui, Z., Li, Y., Igathinathane, C., Flores, P., ... & Zhang, M. (2024). Comprehensive wheat lodging detection after initial lodging using UAV RGB images.Expert Systems with Applications,238, 121788,https://doi.org/10.1016/j.eswa.2023.121788.

[2] C.J. Baker, P.M. Berry, J.H. Spink, R. Sylvester-Bradley, J.M. Griffin, R.K. Scott, R.W. Clare,A Method for the Assessment of the Risk of Wheat Lodging,Journal of Theoretical Biology,Volume 194, Issue 4,1998,Pages 587-603,https://doi.org/10.1006/jtbi.1998.0778.

[3] Guo, Y., Zhou, W., Fu, Y. H., Hao, F., Zhang, X., Xu, L., ... & He, Y. (2025). SegNeXt-RCMSCA: An improved SegNeXt network for detecting winter wheat lodging from UAS RGB images.Smart Agricultural Technology, 101230. https://doi.org/10.1016/j.smartag.2025.101230.

[4] Yu, J., Cheng, T., Cai, N., Zhou, X. G., Diao, Z., Wang, T., ... & Zhang, D. (2023). Wheat lodging segmentation based on Lstm_PSPNet deep learning network.Drones,7(2), 143.. https://doi.org/10.3390/drones7020143.

[5] Biswal, S., Chatterjee, C., & Mailapalli, D. R. (2023). Damage assessment due to wheat lodging using UAV-based multispectral and thermal imageries.Journal of the Indian society of remote sensing,51(5), 935-948..https://doi.org/10.1007/s12524-023-01680-6.

[6] Celik, M. F., Nagarajan, P., Nelson, A., Unnisa, Z., Ogutu, B., Dash, J., ... & Darvishzadeh, R. (2026). Hyperspectral assessment of wheat lodging: From field to EnMAP satellite observations. International Journal of Applied Earth Observation and Geoinformation,149, 105289..https://doi.org/10.1016/j.jag.2026.105289.

[7] Long, J., Zhang, Z., Zhang, Q., Zhao, X., Igathinathane, C., Xing, J., ... & Zhang, M. (2025). Comprehensive wheat lodging detection under different UAV heights using machine/deep learning models. Computers and Electronics in Agriculture,231, 109972. https://doi.org/10.1016/j.compag.2025.109972.

[8] Feng, G., Wang, C., Wang, A., Gao, Y., Zhou, Y., Huang, S., & Luo, B. (2024). Segmentation of wheat lodging areas from UAV imagery using an ultra-lightweight network.Agriculture,14(2), 244. https://doi.org/10.3390/agriculture14020244.

[9] Fan, X., Zhi, S., Tjahjadi, T., Liu, J., Bai, Z., Ye, Q., & Huan, H. (2026). Integrating UAV-acquired RGB and LiDAR point cloud data using multimodal learning for wheat lodging mapping and area estimation.IEEE Journal of Selected Topics in Signal Processing.https://doi.org/10.1109/JSTSP.2026.3156789.

[10] Zhang, P., Zhang, S., Wang, J., & Sun, X. (2024). Identifying rice lodging based on semantic segmentation architecture optimization with UAV remote sensing imaging.Computers and Electronics in Agriculture,227, 109570.https://doi.org/10.1016/j.compag.2024.109570.

[11] Rabieyan, E., Darvishzadeh, R., & Alipour, H. (2024). Correction: Identifcation and estimation of lodging in bread wheat genotypes using machine learning predictive algorithms.Plant Methods,20(1), 86.https://doi.org/10.1186/s13007-024-01203-5.

[12] Tang, Z., Zhu, Y., Huang, M., & Zhu, J. (2024), December). A study on a wheat image recognition system based on transformer and mamba networks with light enhancement. In2024 5th International Conference on Computers and Artificial Intelligence Technology (CAIT)(pp. 111-115). IEEE.https://doi.org/10.1109/CAIT63228.2024.10876542.

[13] Sun, Q., Chen, L., Xu, X., Gu, X., Hu, X., Yang, F., & Pan, Y. (2022). A new comprehensive index for monitoring maize lodging severity using UAV-based multi-spectral imagery. Computers and Electronics in Agriculture,202, 107362.https://doi.org/10.1016/j.compag.2022.107362.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:02:25 UTC from IEEE Xplore. Restrictions apply.